intelligence, surveillance, and reconnaissance (ISR) feeds, through malware. US EA-18G Growler electronic warfare aircraft launched from the *Gerald Ford* and EP-3s from Japan also found their systems jammed, leaving US commanders reliant on incomplete information from satellites and submarines. This significantly slowed the progress of the US flotilla toward the Spratlys and was only partially countered by launching dozens of line-of-sight communications transmitters tethered to medium-altitude balloons from US Navy ships scattered throughout the theater of operations. With the advent of electronic warfare, US policy makers began to fear that a future wave of cyber-attacks would widen the field of conflict to civilian systems, forcing a major US response. For the time being, however, the Chinese hesitated to escalate the crisis horizontally by targeting noncombatants, focusing instead on crippling US operations in the theater of combat.

The Battle of Santiago Island

On September 20, the US flotilla engaged in hostilities with the South Sea Fleet off Santiago Island, near the town of Bolinao on Luzon. The *Gerald Ford* sent F-18s and F-35s for air cover, easily shooting down PLAAF and PLANAF planes from the Spratlys; given the roughly 600-mile distance from PLANAF bases on Hainan Island and PLAAF bases near Guangzhou, and the lack of in-flight aerial refueling, the Chinese were limited to just the three dozen or so fighters on various Spratly bases. Once again, however, HQ-9s launched from Type 052Ds scored hits on US planes, destroying three F-18s. From nearly 100 nm away, two Chinese Type 052C destroyers loosed a volley of six antiship missiles, which scored hits on the *Antietam* and one DDG, disabling both. US naval forces returned fire, sinking one Chinese destroyer and inflicting serious damage on another. The remaining two US DDGs then endured the missile volleys of ten Type 22 missile boats, each of which fired its complement of eight antiship subsonic cruise missiles. While five of the ten Chinese missile boats were destroyed by return American fire, one remaining US DDG was sunk and the other disabled.